

Course Code: BCE-O544
Course Name: INTRODUCTION TO AI

MM: 100 Time: 3 Hr. L T P 3 0 0	Sessional:30 ESE:70 Credit :3
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Prerequisites:	Probability and statistics, Automata and languages
Objectives:	• Provide the most fundamental knowledge to the students so that they can understand what the AI is. • eliminate theoretic proofs and formal notations as far as possible, so that the students can get the full picture of AI easily. • Students who become interested in AI may go on to the graduate school for further study.

NOTE:	The question paper shall consist of two sections A and B. Section A contains 10 short type questions of 6 marks each and student shall be required to attempt any five questions. Section B contains 8 long type questions of ten marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus.
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Module	Course Content	No. of Hours	POs mapped	PSOs mapped
Module-1	Introduction: Introduction to Artificial Intelligence, Foundations and History of Artificial Intelligence, Applications of Artificial Intelligence, Intelligent Agents, Structure of Intelligent Agents. Computer vision, Natural Language Processing.	08	PO1/ PO2	PSO1/ PSO2
Module-2	Introduction to Search: Searching for solutions, Uniformed search strategies, Informed search strategies, Local search algorithms and optimistic problems, Adversarial Search, Search for games, Alpha - Beta pruning.	08	PO2/ PO3	PSO1/ PSO2
Module-3	Knowledge Representation & Reasoning: Propositional logic, Theory of first order logic, Inference in First order logic, Forward & Backward chaining, Resolution, Probabilistic reasoning, Utility theory, Hidden Markov Models (HMM), Bayesian Networks.	08	PO1/ PO2/ PO4	PSO1/ PSO2
Module-4	Machine Learning: Supervised and unsupervised learning, Decision trees, Statistical learning models, Learning with complete data - Naive Bayes models. Expert System: Existing Systems (DENDRAL, MYCIN) domain exploration Meta Knowledge, Self-Explaining System	08	PO2/ PO3	PSO1/ PSO2
Module-5	Capstone Project: Choose a real-world Problem and develop an AI solution end-to-end, Build the AI Model / Use API, Create a User Interface: Web App / Android App/iOS App, Deploy AI App, Documentation: Problem, model, results, future scope, Live demo.	08	PO1/ PO3/ PO4	PSO1/ PSO2
Total No. of Hours		40		

Learning Outcomes:	• Understand AI's fundamental concepts and methods • Acquire knowledge of modern AI tools, including Deep Learning framework TensorFlow and Deep Learning capabilities of RapidMiner. • Learn how to apply AI-based methods to solving practical business problems • Understand implications of AI for business strategies • Examine where the AI technologies are heading within the next few years.
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Suggested books:

S. No.	Name of Authors /Books /Publisher/Year
1.	Rich & Knight, Artificial Intelligence
2.	Stuart Russell, Peter Norvig, "Artificial Intelligence – A Modern Approach", Pearson Education
3.	Charnick, Introduction to A.I., Addison Wesley
4.	Winston, LISP, Addison Wesley
5.	Marcellous, Expert System Programming, PHI
6.	Elamie, Artificial Intelligence, Academic Press
7.	Lloyed, Foundation of Logic Processing, Springer Verlag

CO-PO/PSO MAPPING														
Course Outcomes (COs)	Program Outcomes (POs)												Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	✓	✓											✓	✓
CO2	✓	✓	✓										✓	✓
CO3	✓	✓	✓	✓									✓	✓
CO4	✓	✓		✓									✓	✓
CO5	✓	✓	✓	✓									✓	✓